

ANISH SHILPAKAR

+9779861208287 | anishshilpakar8@gmail.com | [Portfolio](#)

 [anish-shilpakar13](#) |  [juju2181](#)

Bhaktapur, Bagmati - 44800, Nepal

RESEARCH PROFILE

Computer engineering graduate with 3+ years experience building AI and data systems across US healthcare and banking, alongside independent research in speech recognition and NLP for low-resource languages. My research interests lie in deep learning, computer vision, digital twins, multimodal AI, natural language processing, causal ai, and applied data science.

EDUCATION

- **Khwopa College of Engineering(KhCE) (T.U)** Nov 2018 - Mar 2023
Bachelor of Engineering(B.E.) in Computer Engineering
◦ **Percentage:** 84.33% (Distinction)
◦ **Relevant Coursework:** Artificial Intelligence, Machine Learning, Big Data, Data Mining, Data Structures & Algorithms, Database Management Systems, Digital Signal Processing, Computer Networks, Software Engineering, Project Management. Libali, Bhaktapur

TEST SCORES

- IELTS: 8.5 (L - 9, R - 9, W - 7.5, S - 7.5)

EXPERIENCE

- **Leapfrog Technology Inc. ** July 2025 - Present
Senior Software Engineer - Data/AI Kathmandu, Nepal
 - Contributed to the ongoing development of Appealio, Docvocate's AI-driven denials management platform, extending and optimizing the 835/837 claims data pipeline with new features, performance improvements, and bug fixes, while building interactive dashboards that improve visibility and workflow efficiency for both internal developers and end customers.
 - Architected and deployed production-grade ETL pipelines on AWS (Python, PostgreSQL) for US healthcare datasets encompassing insurance claims, payments, and appeals, ensuring fault-tolerant, high-availability data flow at scale.
 - Engineered a hybrid recommendation system for automated payer rejection resolution by applying anomaly detection, similarity search, and unsupervised clustering on healthcare claims data, significantly reducing manual intervention by the customer support team on insurance claim auto-rejections.
 - Integrated large language models (LLMs) into automated data analysis and reporting workflows, enabling AI-driven insight generation and natural language querying on structured healthcare claims data.
 - Designed and developed a Retrieval-Augmented Generation (RAG) based application for AWS certification preparation, leveraging vector similarity search and LLM prompting; adopted by fellow engineers for structured exam preparation.
- **Khwopa College of Engineering ** January 2025 - Present
Project Supervisor Bhaktapur, Nepal
 - Supervising undergraduate capstone and minor projects in deep learning, NLP and computer vision, providing end-to-end mentorship across research planning, model development, experimental evaluation, and technical documentation.
 - Currently supervising and collaborating with a student team on the development of a multi-branch AI-generated image detector, integrating CLIP for semantic representation, SAM 2 for boundary and spatial artifact analysis, and a spectral autoencoder for frequency-domain feature extraction, with cross-attention-based multimodal fusion for robust synthetic-image detection (Capstone Project, 2026–2027).
 - Supervised a student research project comparing cascaded and direct speech-to-text translation architectures for low-resource Nepali, directing experimental design, capacity-matched benchmarking, evaluation of Whisper-large-v3 and MMS, and adaptation of speech recognition models for Nepali-English code-switched speech (Minor Project, May 2026 – September 2026).
 - Guided a student team in designing and implementing a Nepali Speech Recognition and Summarization System using wav2vec2 and Transformer-based architectures, overseeing model fine-tuning, evaluation, and optimization for low-resource Nepali language data (Capstone Project, 2025-2026).
 - Supervised a student team developing a Nepali Handwritten Digit Recognition system (OCR) using TrOCR and Vision Transformers, directing literature review, dataset curation, model training, and iterative performance benchmarking (Minor Project, January 2026 – May 2026).

• **Dlytica Inc.** [🌐]

December 2023 - July 2025

Data Engineer II / Data Architect / Team Lead

Kathmandu, Nepal

- Architected and deployed DataNature, a production-scale open-source data platform for telecom and banking clients (Ooredoo Palestine, Nepali banks), integrating Apache Spark, Airbyte, Trino, Apache Airflow, and MinIO for end-to-end ETL orchestration, data warehousing, and analytical query optimization across multi-cloud and on-premises environments.
- Designed and led development of AI360 SQL-Agent, a natural language-to-SQL query system for Nepali banking institutions, leveraging Retrieval-Augmented Generation (RAG), large language models (LLMs), and vector databases — enabling non-technical banking staff to query complex financial data through conversational interfaces.
- Built an intelligent deposit recommendation system for Nepali banks using collaborative filtering and LLM-based personalization, delivering context-aware financial product recommendations integrated directly into the AI360 banking chatbot.
- Delivered AI360Studio, a self-serve analytics platform for banking clients incorporating Apache Superset dashboards, optimized data models, and query performance tuning, reducing reporting turnaround for banking analysts.
- Led a cross-functional team of data engineers and scientists across AWS, Azure, and on-premises Kubernetes/OpenShift environments, providing technical mentorship and directing architecture decisions for national and international client deployments.

• **Dlytica Academy** [🌐]

January 2024 - July 2024

Data Science & Engineering Instructor

Remote

- Designed and delivered structured curricula on data engineering, machine learning, and AI fundamentals for undergraduate and early-career students across multiple institutions.
- Conducted intensive bootcamps and hands-on workshops at engineering colleges including Khwopa College of Engineering and Sunway International College, covering end-to-end data pipelines, Python-based ML workflows, and applied AI concepts.
- Developed course syllabi, study plans, and practical assignments aligned with industry tools and frameworks including Python, SQL, Apache Spark, and cloud platforms.
- Mentored students through project-based learning, guiding applied data science and engineering projects from problem formulation to implementation.

• **Genese Solution** [🌐]

May 2023 - November 2023

Associate Data Engineer

Lalitpur, Nepal

- Designed and deployed scalable ETL pipelines across AWS (Glue, S3, Athena, Redshift) and Azure (Data Factory, Databricks, SQL Data Warehouse) for enterprise clients including Cross River Bank, USA.
- Optimized production data pipelines for Cross River Bank's US banking operations, improving data reliability and processing efficiency across large-scale financial datasets.
- Led proof-of-concept development for advanced data architectures and delivered analytical dashboards using Power BI and Amazon QuickSight to support stakeholder decision-making.

• **Code Rush Nepal** [🌐]

November 2022 - April 2023

Data Engineering Apprentice

Lalitpur, Nepal

- Built foundational data engineering projects including web scrapers, stock price notification systems, and end-to-end ETL pipelines with data warehousing solutions on AWS and Azure.
- Earned AWS and Azure data engineering certifications, establishing cloud infrastructure expertise that underpins subsequent professional roles.

FELLOWSHIPS & RESEARCH TRAINING

• **Fusemachines** [🌐]

March 2022 - April 2023

AI Fellow

Remote

- Competitively selected for Fusemachines' intensive AI fellowship program focused on applied machine learning research and development
- Completed structured training in machine learning, deep learning, computer vision, and natural language processing, earning microdegree certifications in each domain.
- Developed applied AI systems including an Earthquake Damage Prediction model and a real-time Mask Detection System, gaining hands-on experience in end-to-end model development, training, and deployment.
- Built foundational expertise in the core ML/DL frameworks that underpin subsequent independent research projects in Nepali speech recognition, OCR, and music source separation.

PROJECTS

• **Optimizing Speech-to-Text Systems for Medical ASR**

Self-initiated Research(In Progress)

Tools: [Python, Pytorch, TorchAudio, Canary-Qwen, Whisper, VAD]

- Investigating the applicability of speech-to-text and automatic speech recognition (ASR) systems in medical domains, with a focus on supporting doctors and healthcare professionals in their clinical workflows.
- Researching and optimizing real-time medical ASR pipelines through the integration of speech recognition models, Voice Activity Detection (VAD), and event-driven queuing mechanisms to enable seamless and responsive voice-based interactions.
- Evaluating the feasibility of medical ASR for Nepali speech, including the exploration and collection of domain-specific Nepali medical speech data to support the development and evaluation of localized ASR systems.

• **Deep Learning-Based Music Source Separation for Newari Traditional Instruments**

Self-initiated Research

Tools: [Python, Pytorch, TorchAudio, U-Net, GAN]



- Developed a deep learning pipeline for separating individual instrument sources from Newari traditional music recordings, a critically under-resourced domain in audio ML research
- Designed and evaluated UNet and GAN-based architectures with ensemble methods, achieving competitive Signal-to-Distortion Ratio (SDR) on curated Newari music dataset
- Presented findings at KHWOPA CEEL 2026 National Conference; awarded Best Paper Award in Computer Science Track
- Abstract published in conference proceedings; full manuscript in preparation for journal submission

• **Automatic Nepali Speech Recognition and Summarization System**

Coursework Project at KhCE

Tools: [Python, Pytorch, Wav2Vec2, Librosa, TorchAudio, CNN-ResNet, mT5 Transformer]



- Fine-tuned wav2vec2 transformer model on Nepali speech corpus, achieving appx. 7% Character Error Rate (CER), establishing a strong baseline for low-resource Nepali ASR.
- Developed and compared custom CNN-ResNet architecture against pre-trained wav2vec2, demonstrating superiority of transfer learning approaches for low-resource language settings.
- Implemented both extractive (TextRank) and abstractive (mT5 Transformer) summarization pipelines for Nepali transcripts, enabling end-to-end speech-to-summary workflow.
- Deployed full-stack application with React frontend and FastAPI backend for real-world accessibility in journalism, legal, and business domains.

• **Automatic Nepali Number Plate Recognition System**

Coursework Project at KhCE

Tools: [Python, Tensorflow, YOLO, CNN, OpenCV, Pillow]



- Designed a three-stage recognition pipeline combining YOLOv4 for license plate detection and character segmentation with CNN-LeNet architecture for character-level recognition on Nepali vehicle plates, achieving an accuracy of 93% on character recognition accuracy.
- Integrated detection and recognition models into a Django web application supporting image and video inputs with automated database logging.
- Piloted system for smart parking implementation at Khwopa College of Engineering, demonstrating real-world deployment viability.

• **RAG-Based Intelligent Learning System for AWS Certification**

Work Project at Leapfrog

Applied GenAI Research

Tools: [Python, RAG, Langchain, Langgraph, Chromadb]

- Designed and developed a RAG-based application with chat and question generation features leveraging vector similarity search and LLM prompting to deliver personalized AWS certification preparation.
- Built knowledge base from AWS documentation and practice materials; implemented semantic search for context-aware question answering and adaptive study recommendations.
- Adopted by fellow engineers at Leapfrog Technology for structured certification preparation, demonstrating practical utility beyond proof-of-concept.

• **Hybrid Payer Rejection Auto-Resolution Engine**

Work Project at Leapfrog

Applied ML Research

Tools: [Python, SQL, K-Means, Cosine Similarity]

- Designed and implemented an ML-driven recommendation system to automate resolution of insurance claim rejections, combining anomaly detection, similarity search, and unsupervised clustering on US healthcare claims data.
- Developed a hybrid recommendation engine integrating cosine similarity search with golden record reference table to match rejection patterns to historical resolution strategies.
- Deployed system reduced manual intervention by customer support team on payer claim auto-rejections, demonstrating measurable operational impact in production healthcare environment.

• **AI360 SQL-Agent: Natural Language Interface for Banking Data**

Work Project at Dlytica

Applied NLP Research

Tools: [Python, SQL, RAG, Langchain]

- Architected a natural language-to-SQL query system for Nepali banking institutions using Retrieval-Augmented Generation (RAG), large language models, and vector databases.

- Enabled non-technical banking staff to query complex financial datasets through conversational interfaces, addressing domain-specific NLP challenges in low-resource banking contexts.
- Integrated SQL-Agent with deposit recommendation system using LLM-based personalization, delivering context-aware financial product recommendations through an AI-powered banking chatbot.
- **End to End AWS ETL Pipeline and Data Warehousing** Personal Data Engineering Project
Tools: [Python, Postgresql, AWS S3, AWS Glue, AWS Athena, AWS Redshift] [🔗]
 - Built end-to-end ETL pipelines extracting data from CSVs, transforming into fact & dimension tables, and loading into a data warehouse using both Python + PostgreSQL and AWS stack (S3, Glue, Athena, Redshift).
 - Designed a star schema with one fact table and four dimension tables, using product info and reviews datasets.
 - Delivered two parallel implementations (local vs cloud) to demonstrate skill in both on-premises and cloud architectures.
- **Unstructured Data Streaming and Processing Using Spark and AWS** Personal Data Engineering Project
Tools: [Python, Spark Streaming, Pyspark, AWS S3, AWS Glue, AWS Athena] [🔗]
 - Developed a real-time data pipeline using Spark Streaming to ingest and process unstructured data (CSV, JSON, TXT).
 - Parsed TXT files with regex to extract relevant fields aligned with the destination schema, and stored processed data in AWS S3 as Parquet files for optimized storage and querying.
 - Created Glue Data Catalog via AWS Glue crawler and loaded data into AWS Glue database and performed data analysis with AWS Athena on structured tables for actionable insights

SKILLS

- **Programming Languages:** Python, SQL, C, C++, JavaScript, Bash, PHP
- **Machine Learning & Deep Learning:** Pytorch, Tensorflow, Keras, Scikit-Learn, Hugging Face Transformers, OpenCV, Librosa
- **Data Science & Generative AI:** Numpy, Pandas, Scipy, Langchain, Langgraph, MCP, Agentic Development, Prompt Engineering, LLM Fine-tuning, RAG Development
- **Big Data & Data Engineering:** Apache Spark, Apache Kafka, Apache Airflow, Apache Hive, Trino, NiFi, Airbyte, dbt, PySpark, Kubeflow
- **Cloud & Infrastructure:** AWS (Glue, S3, Redshift, Athena, SageMaker, Bedrock, EC2), Azure (Data Factory, Synapse Analytics, Databricks, ADLS), GCP, Docker, Kubernetes, Red Hat OpenShift, OpenLDAP, Git, Gitlab, Linux
- **Databases & Storage:** PostgreSQL, MySQL, MongoDB, Oracle, SQL Server, Snowflake, MinIO, Amazon S3, Amazon Redshift, Vector Databases (FAISS, ChromaDB)
- **Visualization & BI:** Apache Superset, Power BI, Tableau, Amazon QuickSight, Matplotlib, Seaborn, Plotly
- **Web Frameworks:** FastAPI, Flask, Django, Laravel, React, Node.js
- **Other Tools & Technologies:** Latex, Overleaf, Thingsboard, Flexsim, Canva, Figma, Adobe XD

HONORS AND AWARDS

- **Best Paper Award in Computer Science Track** Feb 2026
Khwopa College of Engineering & Khwopa Engineering College, Bhaktapur, Nepal
 - Awarded to the best work presented on National Conference on Computer, Electrical, Electronics and Communication Engineering (KHWOPA CEEL 2026).
 - Awarded for work on deep learning-based music source separation for Newari traditional instruments

TEACHING EXPERIENCE

- **Project Supervisor, NLP & Computer Vision** 2025 - Present
Khwopa College of Engineering
- **Database Modelling Trainer, Database & Data Engineering** 2025
Leapfrog Technology
- **Data Tutor, Data Engineering & Data Science** 2024
Dlytica Academy
- **Instructor, Data & AI Bootcamps/Workshops** 2024
Khwopa College of Engineering, Sunway College

SERVICE & LEADERSHIP

- **Hackathon Judge & Mentor**
Hack the Circle - Khwopa College of Engineering
 - Evaluated hackathon ideas, proposals, and technical projects.
 - Mentored participating student teams.
 - **AWS Community Builder — Data Category**
Amazon Web Services
 - Represented Nepal in AWS technical community
 - Co-hosted and participated in community events, meetups, and knowledge sharing sessions.
 - Actively promoted AWS in Nepali colleges and student groups, which increased the participation of students and contributed to growth of AWS tech community in Nepal.
 - **Class Representative**
Khwopa College of Engineering
 - Elected representative for undergraduate cohort across three consecutive academic years.
 - Conducted trainings and knowledge sharing sessions and worked to bridge gap between students and faculty.

2024

2023 - Present

2019 - 2022

CERTIFICATIONS

- **AWS:** AWS Certified Data Engineer Associate
 - **AWS:** AWS Certified Solutions Associate
 - **Google:** Google Cloud Generative AI Fundamentals
 - **AWS:** AWS Certified Cloud Practitioner
 - **Coursera:** Natural Language Processing with Probabilistic Models
 - **AWS, Udacity:** AWS Certified Cloud Practitioner
 - **Fusemachines:** Microdegree in Machine Learning
 - **Coursera:** Natural Language Processing with Classification and Vector Spaces
 - **Google:** Google Cloud Essentials
 - **Datacamp:** Python Data Fundamentals

Dec 2025

Apr 2024

Sept 2023

Feb 2023

Dec 2022

Oct 2022

July 2022

July 2022

May 2022

Mar 2022

ADDITIONAL INFORMATION

Languages: English (Fluent), Nepali (Fluent), Hindi (Fluent)